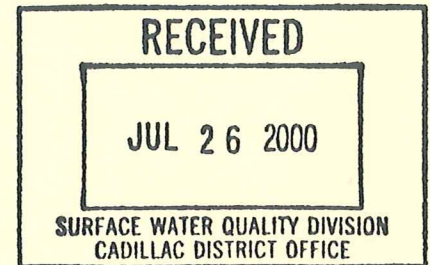




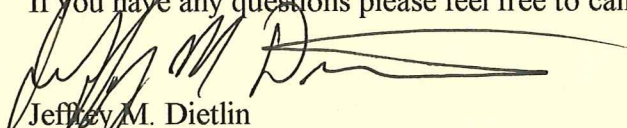
July 26 , 2000

Mr. Mike Person
Michigan Department of Environmental Quality
Cadillac District Office
120 West Chapin Street
Cadillac , Michigan 49601



Enclosed is the RMP report for biosolids that was due on July 7 , 2000. I hope everything is in order. If you need any additional information please contact me.

If you have any questions please feel free to call me.


Jeffrey M. Dietlin
Water Resources Supervisor
Phone # : (231) 775 - 2841



State of Michigan Biosolids Land Application Program

Facility Name
City of Cadillac Waste Water Treatment Plant

Residuals Management Program Development Document

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**SURFACE WATER QUALITY DIVISION
MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY**

PURPOSE

The Michigan Biosolids Program establishes standards for the land application and beneficial recycling of biosolids in the state. Any treatment works treating domestic sewage proposing to land apply biosolids in the state shall prepare the enclosed Residuals Management Program (RMP) Guidance Document to obtain authorization. Authorization is required under the auspices of either an individual NPDES Permit or General Permit "Certificate of Coverage" (COC), dependent upon the type of permit held by the generating facility proposing to land apply.

For assistance on completing any section in this document you may contact the appropriate DEQ Surface Water Quality Division (SWQD) Office listed in Appendix A.

Industrial or commercial facilities with process wastewater entering the waste stream do not qualify for issuance of a biosolids land application permit. Contact the DEQ Waste Management Division in the appropriate district office for information on handling commercial or industrial wastes.

AUTHORITY

The Part 24 Rules, Land Application of Biosolids, of Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended, (NREPA) provide authority to issue Biosolids Land Application Permits. Completion and subsequent approval of this document meets the RMP requirements contained in Rule 2403, Land Application Permit of the Part 24 Rules. Failure to prepare and submit an RMP prior to land applying biosolids is a violation of Part 24 Rules and may result in penalties for such violation.

Rule 2403.

- (1) A generator or distributor shall have a valid permit before commencing any biosolids land application or distribution activity in the state of Michigan.
- (2) A generator or distributor shall submit a permit application on a form provided by the DEQ not less than 180 days before expiration of an existing permit, commencement of biosolids land application, or distribution of a biosolids derivative, except at retail.
- (3) All of the following provisions apply to a residuals management program:
 - (a) A generator shall submit a residuals management program for approval by the DEQ as required by its permit.
 - (b) A residuals management program submitted to the DEQ shall include all of the following information:
 - (i) Size and type of generating facility.
 - (ii) One year of records representing the volume and concentrations of pollutants in the biosolids.
 - (iii) Treatment process origin, for example, primary or secondary treatment and the volume of biosolids generated from each process.
 - (iv) A description of the treatment processes.
 - (v) Storage volume.
 - (vi) Transportation methods and spill prevention plan.
 - (vii) Land application method.
 - (viii) Land application site list.
 - (ix) Land application plan.
 - (x) Pathogen reduction method.
 - (xi) Vector attraction reduction method.
 - (xii) Monitoring program.
 - (c) Upon approval by the DEQ, the generating facility shall implement the approved residuals management program.
 - (d) A generating facility may modify the approved residuals management program by submitting a proposed modification to the DEQ for approval. The modification shall become effective upon approval by the DEQ.
- (4) A person shall land apply biosolids or prepare biosolids for land application in accordance with the requirements established in these rules.

The Michigan Department of Environmental Quality (DEQ) will not discriminate against any individual or group on the basis of race, sex, religion, age, national origin, color, marital status, disability, or political beliefs. Questions or concerns should be directed to the Office of Personnel Services, PO Box 30473, Lansing MI 48909

RESIDUALS MANAGEMENT PROGRAM

SECTION III – LAND APPLICATION PLAN

Instructions for completing Section III.

The information provided in this Land Application Plan section outlines the generating facility's RMP. In addition to requirements set forth in the Michigan Part 24 Biosolids Rules, the information provided (except items marked **Optional**) represent enforceable commitments to manage the land application program in the manner indicated. If changes in circumstances arise, opportunities do exist to request modification to approved RMPs during the biosolids permit cycle.

Although land application contractors are commonly utilized to handle many operational aspects of land application programs, each facility is required to take an active role in managing and overseeing certain administrative aspects of their land application program. Although the DEQ may hold contractors responsible for certain violations of Part 24 rules, the generating facility is responsible for ensuring compliance with state and federal statutes and the conditions found in the approved RMP.

- A. **Biosolids Treatment** - Identify the exact alternatives used at the facility to meet Pathogen Reduction and Vector Attraction Reduction requirements and include the appropriate documentation to demonstrate compliance with those conditions. If your facility utilizes more than one alternative to meet the requirements, indicate all that apply. Provide additional information on attached sheets.
- B. **Procedures** – General programmatic information outlining the facilities biosolids operating procedures.
- C. **Site Information** – Information specific to your land application sites and overall site management.

RESIDUALS MANAGEMENT PROGRAM

SECTION III – LAND APPLICATION PLAN

The following Site Restrictions contained in R323.2414 (f)(I) to (viii) apply to all land-applied biosolids meeting the class B pathogen reduction requirements.

(f) All of the following provisions apply to site restrictions:

- (i) A land owner shall not harvest food crops that have harvested parts which touch the biosolids/soil mixture and which are totally above the land surface for 14 months after biosolids are applied.
- (ii) A land owner shall not harvest food crops that have harvested parts below the surface of the land for 20 months after biosolids are applied if the biosolids remain on the land surface for 4 months or longer before incorporation into the soil.
- (iii) A land owner shall not harvest food crops that have harvested parts below the surface of the land for 38 months after biosolids are applied if the biosolids remain on the land surface for less than 4 months before incorporation into the soil.
- (iv) A land owner shall not harvest food crops, feed crops, and fiber crops for 30 days after biosolids are applied.
- (v) A land owner shall not graze animals on the land for 30 days after biosolids are applied
- (vi) A land owner shall not harvest turf grown on land where biosolids are applied for 1 year after biosolids are applied if the harvested turf is placed on either land that has a high potential for public exposure or a lawn, unless otherwise specified by the permitting authority.
- (vii) A land owner shall restrict public access to land that has a high potential for public exposure for 1 year after biosolids are applied.
- (viii) A land owner shall restrict public access to land with a low potential for public exposure for 30 days after biosolids are applied.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****Minimum Sample Frequencies****R 323.2412 Frequency of monitoring.**

Table 7

Amount of Biosolids Produced (per 365-day period)		
English Dry Tons	Metric Dry Tons	Frequency
Greater than zero, but less than 320	Greater than zero, but less than 290	Once per year
Equal to or greater than 320, but less than 1,650	Equal to or greater than 290, but less than 1,500	Once per quarter (4 times per year)
Equal to or greater than 1,650, but less than 16,500	Equal to or greater than 1,500, but less than 15,000	Once per 60 days (6 times per year)
Equal to or greater than 16,500	Equal to or greater than 15,000	Once per month (12 times per year)

Note: Additional monitoring beyond that stated above may be required to obtain representative data of the material proposed to be land applied or to meet the following additional requirements of the Part 24 Rules. The following words in bold below are not part of the Part 24 Rules.

(4) If biosolids are accumulated before removal, the monitoring frequency, at a minimum, is that specified in this rule. If monitoring of biosolids or a derivative indicates a pollutant concentration in excess of that provided in table 3 of R 323.2409(5)(c) (**see page 11 of this document**), then the monitoring frequency shall be increased to not less than twice that provided for in table 7 (**sample frequency table above**) until pollutant concentrations are at or below the concentrations provided in table 3 of R 323.2409(5)(c). In the case of biosolids accumulating for periods of more than 1 year, biosolids must be monitored at the frequency determined in table 7 only in the year the biosolids are used. For a generator who removes biosolids monthly or more frequently, monitoring is required at least once per month each month that the biosolids are removed, unless more frequent monitoring is required in subrule(1) of this rule.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****R 323.2409 APPLICATION OF LIMITATIONS**

(a) TABLE 1 -- Ceiling Pollutant Concentrations

Pollutant	Ceiling Concentration (milligrams per kilogram) ¹ (dry weight basis)
Arsenic	75
Cadmium	85
Copper	4300
Lead	840
Mercury	57
Molybdenum	75
Nickel	420
Selenium	100
Zinc	7500

(b) TABLE 2 -- Cumulative Pollutant Loading Rates

Pollutant	Cumulative Pollutant Loading Rate	
	kilograms per hectare	pounds per acre
Arsenic	41	37
Cadmium	39	35
Copper	1500	1335
Lead	300	267
Mercury	17	15
Nickel	420	374
Selenium	100	89
Zinc	2800	2492

(c) TABLE 3 -- Pollutant Concentrations

Pollutant	Concentration (milligrams per kilogram) (on a dry weight basis)
Arsenic	41
Cadmium	39
Copper	1500
Lead	300
Mercury	17
Nickel	420
Selenium	100
Zinc	2800

(d) TABLE 4 -- Annual Pollutant Loading Rates

Pollutant	Annual Pollutant Loading Rate ¹	
	kilograms per hectare	pounds per acre
Arsenic	2.0	1.8
Cadmium	1.9	1.7
Copper	75	67
Lead	15	13
Mercury	0.85	0.76
Nickel	21	19
Selenium	5.0	4.5
Zinc	140	125

¹ per 365-day period.

RESIDUALS MANAGEMENT PROGRAM**SECTION III – LAND APPLICATION PLAN****MANAGEMENT PRACTICES****Slope Restrictions R 323.2410(4)**

A person shall not apply bulk biosolids on lands having a slope of more than 6% for surface application or more than 12% for subsurface injected biosolids, unless the person uses the bulk biosolids in accordance with a DEQ-approved site management plan.

Isolation Distance Requirements R 323.2410(11)

TABLE 6

ISOLATION DISTANCE REQUIREMENTS		
Isolation from existing:	Distance (feet)	
	Injection or Surface application with incorporation*	Surface application without incorporation
Municipal well (type I or type IIA)**	2000	2000
Non-community public water supply (type IIB or type III)	800	800
Domestic well	100	150
Homes	100	150
Commercial Buildings	100	150
Surface waters***	50	150

* Incorporation must be within 48 hours, unless a shorter time period is specified in these rules.

** As defined and specified in Act No. 399 of the Public Acts of 1976, as amended, being §325.1001 et seq. of the Michigan Compiled Laws and known as the safe drinking water act. As specified in Act No. 399 of Public Acts of 1976, as amended, the term includes water supplies such as schools, restaurants, industries, campgrounds, parks, and hotels.

*** Surface waters do not include grassed drainage ways or drainage ways that are tilled and planted.

Refer to Rule R 323.2410 for the entire management practices section of Part 24 Rules.